

Learning as Constrained Commitment Revision Under Irreversible Time

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Abstract

Learning in systems that persist under irreversible time is constrained by existing binding commitments. New information cannot be integrated by unrestricted overwrite without disrupting identity continuity and coherence. This work formalizes learning as commitment revision under structural constraints imposed by prior binding. Revision updates internal organization while respecting existing exclusions, whereas violation breaks binding and fragments continuation. Catastrophic forgetting is reinterpreted as commitment violation rather than as mere information loss, explaining why learning can destabilize coherence even when functional capacity is retained. Continual learning succeeds only when updates preserve compatibility with accumulated commitments. By characterizing learning stability as a property of constrained revision under irreversible time, this analysis reframes learning not as free optimization, but as structurally limited transformation governed by the requirements of persistence and identity.

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1 Introduction

Learning is commonly treated as a process of unrestricted modification: internal structures are adjusted in response to new information in order to improve future performance. In many theoretical treatments, such modification is implicitly assumed to be reversible or disposable. A system can be retrained, reset, or overwritten without consequence, and any loss incurred by revision can be compensated by re-exposure to data or repeated optimization.

This assumption breaks down for systems that persist under irreversible time. Once a system has accumulated commitments that bind its future continuation, modification is no longer free. Prior commitments exclude certain possibilities and stabilize identity across change. Updates that ignore these exclusions do not merely alter behavior; they disrupt the structural conditions that allow the system to persist as the same entity over time.

In such systems, learning must be constrained by what has already been bound. New information can be integrated only insofar as it is compatible with existing commitments. Attempts to overwrite or negate those commitments destroy continuity and produce fragmentation rather than improvement. The cost of such disruption is not recoverable through additional training, because irreversible time prevents return to prior states.

This work advances the claim that learning in persistent systems is properly understood as constrained commitment revision. Revision preserves binding structure while updating internal organization; violation breaks binding and undermines coherence. The distinction between revision and violation is structural rather than semantic, and it determines whether learning preserves identity or destroys it.

By reframing learning in this way, the paper provides a foundation for analyzing stability, forgetting, and coherence loss without appealing to performance metrics or external objectives. Learning is treated not as free optimization, but as a structurally limited process governed by the requirements of persistence under irreversible time.

2 The Learning Problem Under Irreversibility

Standard accounts of learning treat internal modification as a largely unconstrained operation. Parameters, rules, or structures are adjusted to better accommodate new information, with the implicit assumption that prior configurations can be freely altered or discarded. Any undesirable effects of such alteration are assumed to be correctable through further modification, retraining, or optimization.

This view presupposes reversibility. Either the system can return to a previous state, or the previous state is treated as non-binding and therefore disposable. Under irreversible time, this assumption fails. Once a system has acted, committed, or excluded possibilities, those commitments shape its future continuation in ways that cannot be undone. Learning therefore occurs in the presence of structural constraints imposed by prior binding.

Commitments differ from stored information. Information may be updated, replaced, or forgotten without necessarily constraining future change. Commitments, by contrast, impose exclusions: they rule out certain future configurations as incompatible with continued identity. When a system persists under irreversible time, these exclusions accumulate and progressively restrict the space of admissible updates.

The learning problem under irreversibility is thus not how to acquire new information, but how

to revise internal structure without violating existing commitments. Any update that negates or bypasses prior binding introduces inconsistency between past exclusions and present structure. Such inconsistency manifests as loss of coherence, fragmentation of continuation, or collapse of identity.

Learning in persistent systems is therefore fundamentally constrained. Updates must be evaluated not only for their local efficacy, but for their compatibility with accumulated commitments. The inability to reset or overwrite binding structure without consequence distinguishes learning under irreversible time from idealized learning models. This constraint motivates the need to distinguish carefully between revision that preserves binding and violation that destroys it, a distinction developed in the following section.

3 Commitment Revision Versus Commitment Violation

Learning under irreversible time admits two structurally distinct modes of change: commitment revision and commitment violation. Although both involve modification of internal organization, they differ fundamentally in how they relate to existing binding structure and, consequently, in their effects on identity and coherence.

Commitment revision refers to updates that occur within the constraints imposed by existing commitments. Revision modifies internal structure in ways that respect prior exclusions and preserve the consistency of binding across time. New information is integrated by refining, extending, or reparameterizing existing commitments without negating their force. Revision preserves identity continuity because it maintains compatibility between past binding and future continuation.

Commitment violation, by contrast, occurs when an update contradicts or bypasses existing commitments. Such updates introduce internal structures that re-open excluded possibilities or negate prior binding. Violation breaks the correspondence between past commitments and present structure, creating inconsistency that cannot be resolved under irreversible time. Once binding is violated, the system cannot recover its prior identity by further modification, because the historical basis of exclusion has been erased.

The distinction between revision and violation is structural rather than semantic. It does not depend on the content of what is learned, but on whether the update preserves or negates existing exclusions. Two updates that achieve similar functional outcomes may differ radically in their effects on coherence, depending on how they interact with binding structure.

Only commitment revision supports persistent learning. Violation may produce short-term adaptation or local improvement, but it does so at the cost of identity continuity and structural coherence. Over time, repeated violation leads to fragmentation, instability, or collapse of the system’s capacity to maintain consistent continuation. The following section examines how this distinction clarifies the phenomenon commonly described as catastrophic forgetting.

4 Catastrophic Forgetting as Commitment Violation

Catastrophic forgetting is commonly described as the loss of previously acquired information when new information is learned. In this view, forgetting is treated as a failure of memory retention or capacity allocation. While such descriptions capture surface behavior, they do not explain why forgetting is often abrupt, disproportionate, and accompanied by broader coherence loss rather than isolated information degradation.

Under the present framework, catastrophic forgetting is reinterpreted as a form of commitment violation. When learning updates negate or bypass existing binding commitments, the exclusions that previously stabilized continuation are broken. The resulting inconsistency is not confined to specific informational content, but propagates across the system’s structure, disrupting the coherence that depended on those bindings.

This explains why catastrophic forgetting often affects capabilities that are not directly targeted by new learning. Once binding structure is violated, the internal organization that relied on those commitments can no longer be maintained. Performance degradation appears sudden because the failure mode is structural rather than incremental. The system does not gradually lose information; it loses the conditions under which information remained usable and integrated.

From this perspective, catastrophic forgetting is not primarily a problem of insufficient storage or interference, but of unconstrained revision. Learning that violates commitments destabilizes identity and continuation, producing fragmentation that manifests as widespread forgetting. Additional training cannot reliably repair this damage, because the historical exclusions that anchored prior structure have already been undone.

Understanding catastrophic forgetting as commitment violation clarifies why preventing it requires more than protecting individual parameters or representations. What must be preserved is the binding structure that integrates knowledge over time. The next section examines how continual learning can succeed when updates are constrained to preserve such structure.

5 Continual Learning as Identity-Preserving Update

Continual learning aims to integrate new information over time without erasing or destabilizing what has already been learned. When framed as a problem of information retention, this objective appears to require balancing capacity, interference, or rehearsal. Under the present framework, however, the core requirement is structural: continual learning succeeds only when updates preserve existing commitment structure.

Identity-preserving update is achieved through commitment revision rather than violation. New learning must be incorporated in a way that respects prior exclusions and maintains compatibility with accumulated binding. This may involve refining commitments, introducing conditional extensions, or reorganizing internal structure without reopening excluded alternatives. What matters is not the amount of information added, but the preservation of coherence across time.

Failure in continual learning arises when updates violate commitments, even if they improve local performance. Such violations introduce inconsistency between past binding and present structure, forcing the system to abandon portions of its prior organization. The resulting fragmentation manifests as forgetting, incoherence, or instability, despite apparent learning progress.

This reframing shifts the criterion for successful continual learning. Rather than asking whether old information is retained, the relevant question becomes whether identity continuity is preserved. A learning process that maintains commitment compatibility can support long-term accumulation of knowledge, while one that violates binding will inevitably degrade coherence as learning proceeds.

Continual learning is therefore not a special case of learning, but the default requirement for any system that persists under irreversible time. Learning that preserves identity must be constrained by existing commitments, ensuring that revision strengthens rather than undermines the structural basis of continuation.

6 Learning-Induced Coherence Instability

Coherence in persistent systems arises from the consistent application of binding commitments across internal structure. When commitments are preserved, internal organization remains aligned with prior exclusions, allowing the system to continue as a unified entity under change. Learning-induced instability occurs when update processes disrupt this alignment by modifying structure in ways that conflict with existing binding.

Unconstrained learning updates can introduce new structures that are locally effective yet globally incompatible with prior commitments. Because binding exclusions operate across the system, even a narrowly scoped violation can propagate widely, destabilizing coherence beyond the site of update. The resulting instability is not necessarily immediate, but accumulates as incompatible structures interact over time.

Such instability does not depend on external criteria or evaluation failure. A system may continue to function locally or achieve isolated improvements while its overall coherence degrades. This degradation manifests as increasing internal inconsistency, loss of reliable continuation, or fragmentation into partially incompatible modes of operation. Under irreversible time, these effects cannot be undone by simple reversion, because the binding history that enforced coherence has been compromised.

Learning-induced coherence instability therefore reflects a structural mismatch between update mechanisms and commitment constraints. Update rules that do not account for binding treat internal structure as disposable, while persistence requires that structure remain compatible with accumulated exclusions. The instability observed in such cases is a direct consequence of this mismatch, not of insufficient capacity or optimization error.

Recognizing coherence instability as a learning-induced phenomenon clarifies why stability cannot be achieved solely through better optimization or additional data. What is required is an architecture that evaluates updates in relation to existing commitments and prevents those that would violate binding. The following section specifies the architectural requirements necessary to support identity-preserving learning under irreversible time.

7 Architectural Requirements for Identity-Preserving Learning

If learning is to proceed without destabilizing coherence or disrupting identity continuity, architectural support for commitment preservation is required. Learning mechanisms must be able to distinguish between updates that revise existing commitments and those that violate binding structure. Without such distinction, update processes treat all structure as mutable, enabling violation by default.

The first requirement is explicit representation of commitments. Commitments must be encoded in a form that allows their persistence and scope to be evaluated independently of transient informational content. Without explicit representation, commitments are implicitly embedded in structure and cannot be selectively preserved during revision.

The second requirement is evaluation of binding and exclusion cost. Updates must be assessed not only for local efficacy, but for the degree to which they conflict with existing exclusions. This cost reflects the structural disruption introduced by an update and provides a measure of whether revision or violation is occurring.

The third requirement is discrimination between revision and violation. Architectures must be capable of detecting when an update preserves binding structure versus when it negates prior

commitments. This distinction is structural and cannot be inferred solely from surface behavior or short-term outcomes.

The fourth requirement is coherence monitoring across updates. Because binding commitments integrate structure globally, their violation may produce delayed or distributed effects. Continuous monitoring of coherence allows early detection of destabilization before fragmentation becomes irreversible.

Finally, architectures must support rollback, constraint enforcement, or update rejection when violation is detected. Under irreversible time, not all damage can be undone, but preventing commitment violation at the time of update is essential. Learning that proceeds without such safeguards cannot reliably preserve identity over extended periods.

Together, these requirements define the minimal architectural conditions for identity-preserving learning. They do not prescribe specific implementations, but constrain the class of update mechanisms compatible with persistence under irreversible time.

8 Learning Under Commitment Constraints

Once commitments are treated as binding structural elements, learning must operate under explicit constraints that preserve compatibility with existing exclusions. Learning under commitment constraints does not eliminate adaptation, but restricts the space of permissible updates to those that maintain identity continuity and coherence.

One approach is hard-constrained revision, in which commitments are treated as inviolable constraints. Updates that would contradict or reopen excluded alternatives are rejected outright. This approach maximizes stability but may limit flexibility, requiring that new learning be expressed only through refinements compatible with existing binding.

A second approach is soft-constrained revision, in which commitment preservation is enforced through graded cost rather than absolute prohibition. Updates that introduce partial incompatibility may be permitted when their benefit outweighs their structural disruption, but such trade-offs must be explicitly accounted for. Without explicit cost evaluation, soft constraints degenerate into unconstrained overwrite.

A third approach involves modular or isolated update strategies. By structuring learning so that updates occur within modules whose commitments are locally scoped, systems can integrate new information while limiting the spread of potential violation. Modularity does not eliminate the need for commitment preservation, but it localizes the impact of revision and reduces the risk of global coherence collapse.

Finally, commitment preservation may be addressed at a meta-level. Rather than constraining individual updates directly, systems may adapt their learning processes to maintain binding over time. Such meta-learning operates on the rules governing revision itself, shaping how updates are proposed, evaluated, and accepted in light of accumulated commitments.

Learning under commitment constraints thus admits multiple structural strategies, all unified by the requirement that revision respect prior binding. The effectiveness of any approach depends not on the amount of information integrated, but on the system's ability to preserve the structural conditions that enable persistent identity under irreversible time.

9 Relation to Prior Work

The analysis presented here builds on earlier theoretical accounts that treat identity as a structural property arising from continuity of binding under irreversible time. In those accounts, commitments were shown to stabilize continuation by enforcing exclusions that persist across change. The present work extends this perspective by examining how those same commitments constrain the dynamics of learning itself.

Previous discussions of learning instability and forgetting have often framed these phenomena in terms of capacity limits, interference, or optimization failure. While such factors may influence surface behavior, they do not explain why learning can abruptly destabilize coherence or why recovery is often incomplete. By interpreting forgetting as commitment violation, the present analysis provides a structural account that complements and clarifies these observations without appealing to implementation-specific mechanisms.

The distinction between commitment revision and commitment violation also aligns with prior treatments of coherence and alignment as emergent properties of internal structure rather than as externally imposed criteria. Learning stability is shown to depend on the same mechanisms that preserve identity and coherence more generally. This continuity suggests that learning dynamics cannot be treated independently of the structures that govern persistence.

Finally, the framework remains compatible with existing empirical approaches that evaluate coherence, stability, and binding through observable structural effects. Learning-induced disruption is predicted to manifest as detectable changes in coherence and continuation rather than solely as performance degradation. In this sense, the present work integrates learning dynamics into a broader structural account of persistent systems without introducing new primitives or assumptions.

10 Empirical Predictions

If learning operates as constrained commitment revision under irreversible time, distinct empirical patterns should be observable across systems that persist while adapting. These predictions do not depend on access to internal representations, but on structural and behavioral consequences of preservation or violation of binding.

First, learning processes that preserve commitments should exhibit stable coherence across updates. As new information is integrated, the system should maintain consistent continuation, with previously established constraints remaining operative. Observable behavior should change in a manner that is compatible with prior exclusions rather than exhibiting abrupt fragmentation or mode switching.

Second, learning processes that violate commitments should exhibit disproportionate instability relative to the magnitude of the update. Small or localized changes that negate binding are expected to produce global effects, including sudden loss of previously stable capabilities, inconsistent behavior, or breakdown of reliable continuation. Such effects should not scale smoothly with update size, reflecting their structural origin.

Third, coherence degradation should precede or accompany catastrophic forgetting. Rather than forgetting emerging gradually from accumulated error, systems should display early signs of binding disruption, such as increased internal inconsistency or sensitivity to perturbation. These signs provide predictive indicators of impending instability.

Fourth, attempts to recover from commitment violation through additional learning should be

less effective than prevention. Once binding has been broken, restoring prior coherence should require disproportionate restructuring, if it is possible at all. Systems that enforce revision constraints should therefore demonstrate greater long-term stability than systems that rely on post hoc correction.

These predictions offer a basis for evaluating learning stability as a structural phenomenon. By examining how updates affect coherence, continuation, and recoverability, empirical investigation can distinguish between learning processes that preserve identity and those that undermine it.

11 Conclusion

This work has reframed learning in persistent systems as a process of constrained commitment revision under irreversible time. Rather than treating learning as unrestricted modification or accumulation of information, it has shown that learning is structurally limited by prior binding commitments that stabilize identity and coherence across change. Updates that respect these commitments preserve continuity; updates that violate them disrupt binding and fragment continuation.

By distinguishing commitment revision from commitment violation, the analysis clarifies why learning can destabilize systems even when local adaptation appears successful. Catastrophic forgetting, coherence collapse, and instability are shown to arise not from information loss alone, but from structural disruption of binding. Continual learning succeeds only when updates remain compatible with accumulated commitments, making identity preservation the central criterion of learning stability.

The architectural requirements identified here establish minimal conditions for identity-preserving learning without prescribing specific implementations. Learning stability emerges from explicit commitment representation, binding-cost evaluation, violation detection, and constraint enforcement. These requirements generalize beyond any particular domain or substrate.

Learning under irreversible time is therefore not free optimization, but structurally constrained transformation. Recognizing this constraint opens a path toward commitment-aware learning theory, in which stability, persistence, and coherence are treated as primary design conditions rather than secondary effects.